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PROFITABILITY UNDER CARBON PRICING AND EXPLORATION RISK

Chair of Energy Systems & Energy Economics

A Techno-Economic Comparison of Offshore Oil and Offshore Wind-Based Green Hydrogen

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AGENDA

1. Motivation & research question
2. Methodology
3. Data & key assumptions
4. Results & discussion
5. Conclusion

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MOTIVATION | WHY DOES CAPITAL STILL PREFER OIL?

€500 bn / year
into oil upstream
30 % chance of finding anything

≈ zero FIDs
into offshore green H_2
no resource risk at all

RESEARCH QUESTION

Same €3 bn — oil or green hydrogen?

under carbon pricing, exploration risk and cost uncertainty
Gap: always studied separately

CONTRIBUTION

- **Capital-matched** — one framework
- **Exploration risk** — made explicit
- **Carbon at two boundaries** — ETS vs. Scope 3

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METHODOLOGY

TWO CAPITAL-MATCHED OFFSHORE PROJECTS

≈ €3 bn CAPEX each · identical 25-year horizon · same carbon overlay

MODEL A

Offshore oil

deepwater, 1,200 m (Qiu et al., 2022)

€3,067M CAPEX

10 % WACC

378 Mbbl · 55,000 bbl/d plateau

ETS: 17 kg CO_2/bbl at €65/t, +7 %/a

MODEL B

Wind → hydrogen

German Bight, 500 MW + 400 MW PEM

€2,938M CAPEX

7 % WACC

33,976 t H_2/a · 80 km pipeline

Offtake €5.50/kg — tested vs. LCOH

Symmetric boundaries: decommissioning both sides · clipping losses

Energy matching would need > 12 GW of wind.

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UNCERTAINTY, CARBON PRICING AND EXPLORATION RISK

Symmetric where possible — asymmetric where reality is asymmetric

UNCERTAINTY

Monte Carlo

triangular distributions
7 oil / 9 H_2 parameters
+ tornado sensitivities

POLICY

Carbon scenarios

EU ETS €0–400/t
2030 central: €120/t
Scope 3 stress: 450 kg/bbl

GEOLOGY

Exploration gate

oil only
PoS = 30 %
sunk E&A = €80M

$$E[NPV] = PoS \times NPV|success - C(E\&A)$$

Fiscal stress: North-Sea-style 65 % petroleum tax
Gate ignores abandonment → understates oil → conclusion is conservative.

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DATA & KEY ASSUMPTIONS

Key assumptions — condensed from paper Tables 1–2

Parameter	Model	Base	MC range	Source
Oil price (€/bbl)	A	45	30–70	Qiu et al. (2022)
Recovery factor	A	0.33	0.11–0.64	Qiu et al. (2022)
ETS price (€/t), escal. 7 %/a	A	65	50–200	European Commission (2025)
PoS / E&A cost	A	30 % / €80M	0.20–0.40	Westwood (2024)
Electrolyser CAPEX (€/kW)	B	2,800	1,000–3,500	IEA (2024)
H_2 offtake price (€/kg)	B	5.50	2.5–10	IEA (2022); Travaglini et al. (2024)
Wind capacity factor	B	0.46	0.38–0.54	IRENA (2024); NREL (2024)
WACC (oil / H_2)	**A / B**	**10 % / 7 %**	**H_2: 3–15 %**	**Qiu et al. (2022); OECD (2024)**

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RESULTS

BASE-CASE RESULTS AND RISK METRICS

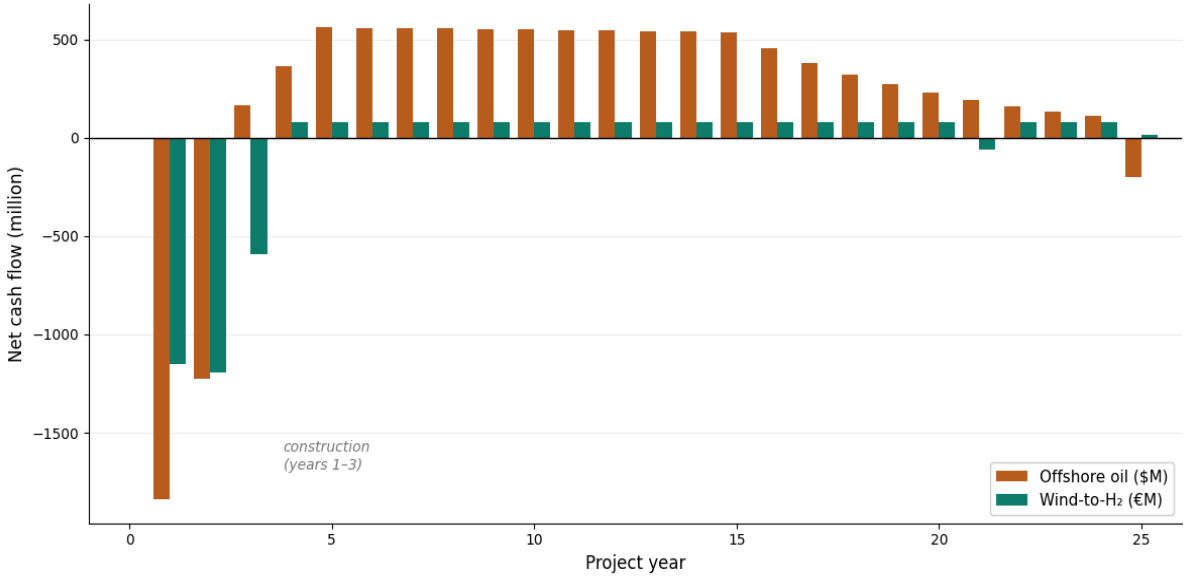
Base-case results and Monte Carlo risk metrics (paper Table 3) — all values in €, EUR/USD parity 1.0

Metric	Offshore oil	Wind-to-H_2 (PEM)
Total CAPEX	€3,067M	€2,938M
NPV (base)	€502M	−€1,909M
IRR	12.4 %	−4.8 %
Breakeven / LCOH	€39.7/bbl	€10.96/kg
MC mean NPV (3,000 runs) *	€2,682M	−€1,855M
P(NPV > 0), conditional	91 %	1 %
P(NPV > 0), unconditional	27 %	1 % (no resource risk)
E[NPV] incl. exploration gate	**€71M**	**−€1,909M**

** Simulation approximates revenue by reserves × price — lies above the DCF value (paper Sec. 2.5). Oil converted at the EUR/USD parity of 1.0 assumed throughout the paper.*

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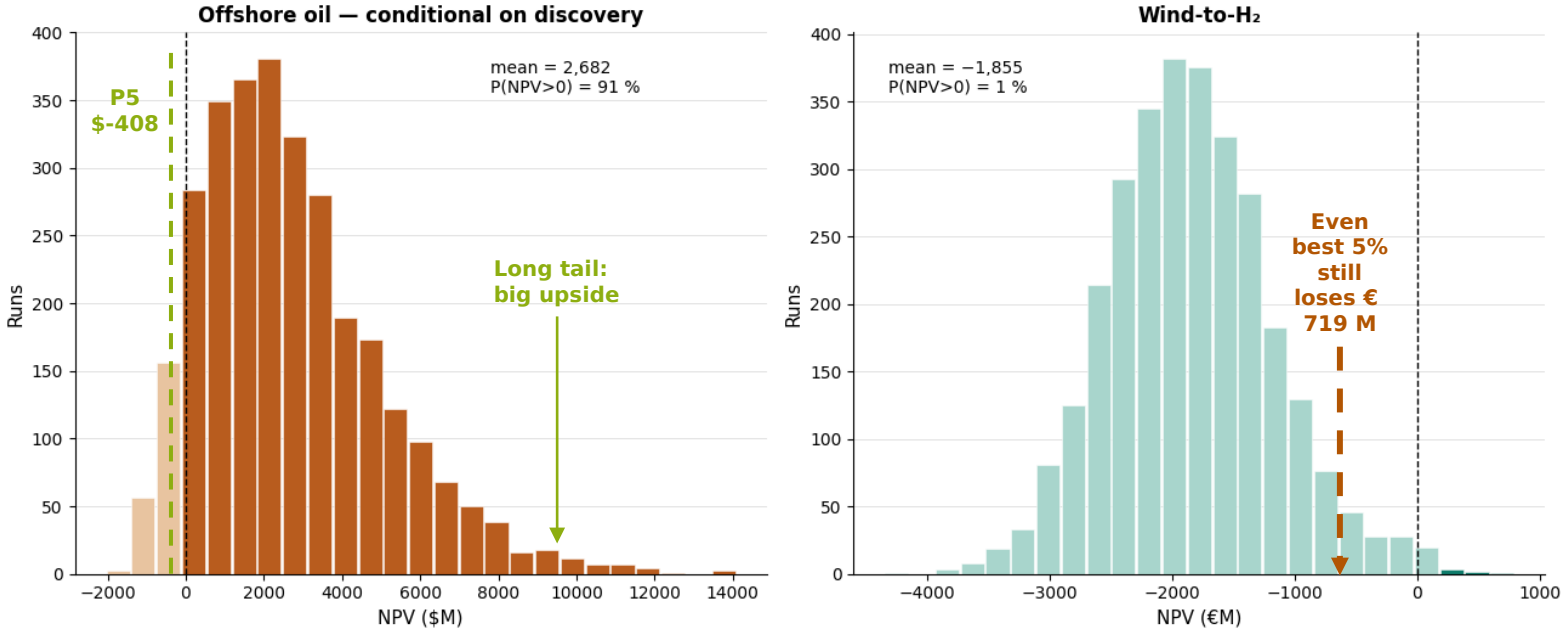
CASH FLOW PROFILES — TWO DIFFERENT WORLDS



Oil: two heavy construction years, then high declining surpluses · H_2: three construction years, then 22 small flat surpluses (Fig. 1)

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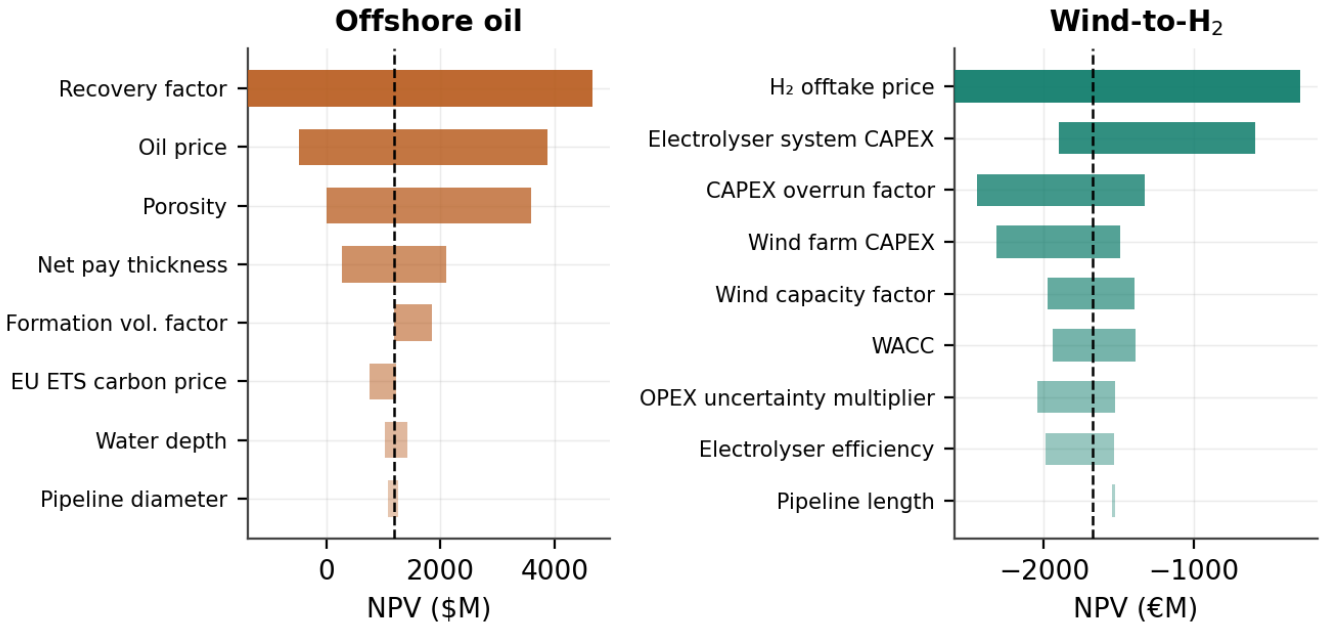
MONTE CARLO NPV DISTRIBUTIONS



Uncertainty is not hydrogen's problem — the central estimate is: even the 95th percentile stays at −€719M (Fig. 2)

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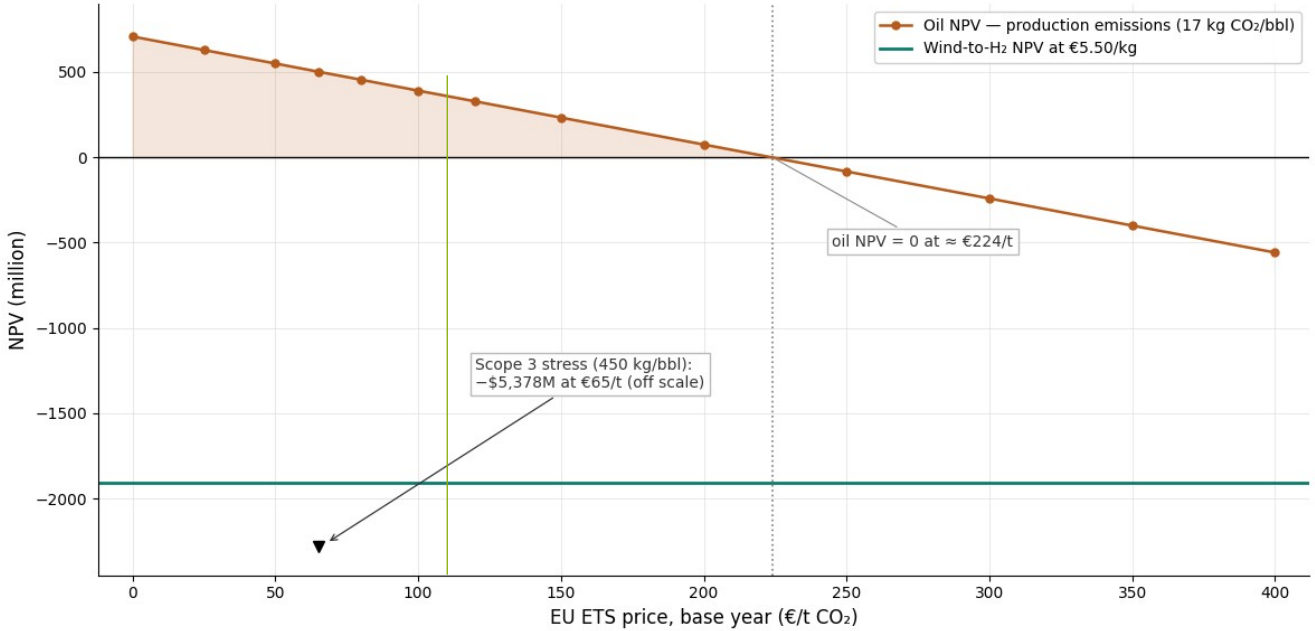
VALUE DRIVERS: TORNADO ANALYSIS



Oil is driven by geology and price — the ETS swing stays below €0.5 bn · H_2 by offtake price, CAPEX and WACC (Fig. 3)

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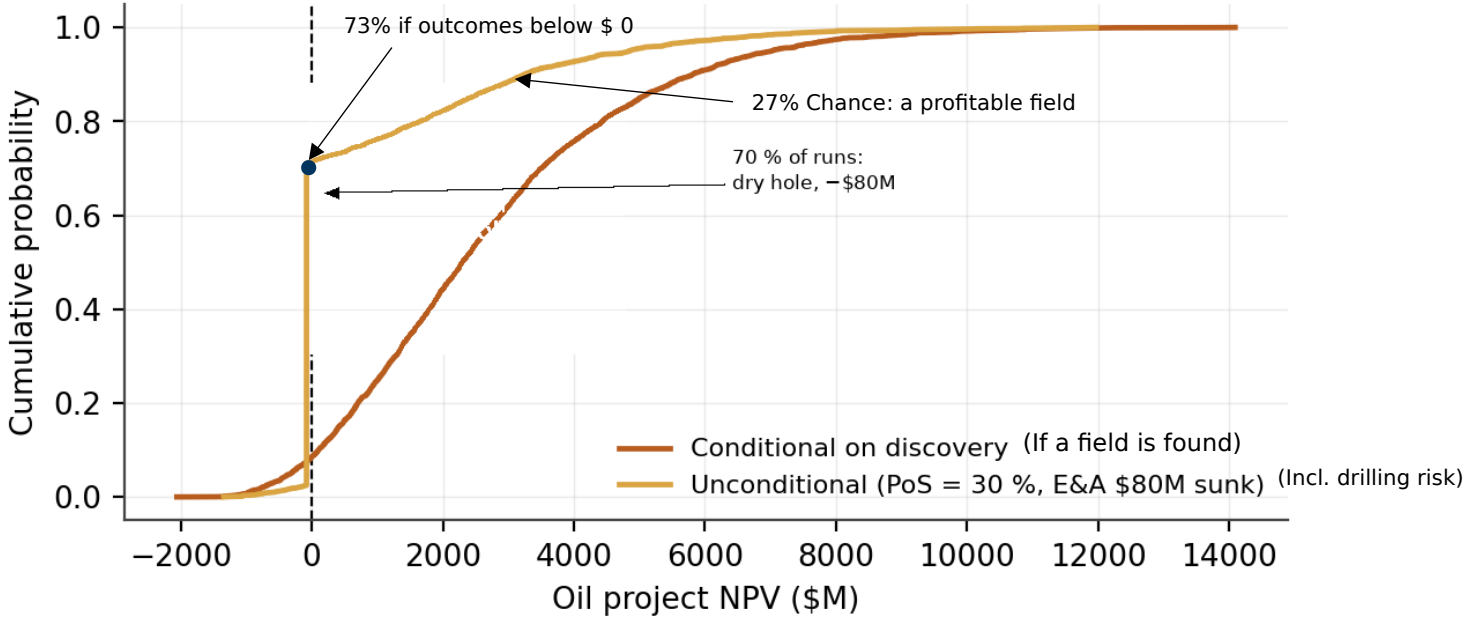
CARBON PRICING AND THE SCOPE 3 BOUNDARY



Oil stays NPV-positive up to ≈ €224/t — ≈ 96 % of lifecycle emissions sit outside the producer's ETS liability (Fig. 4)

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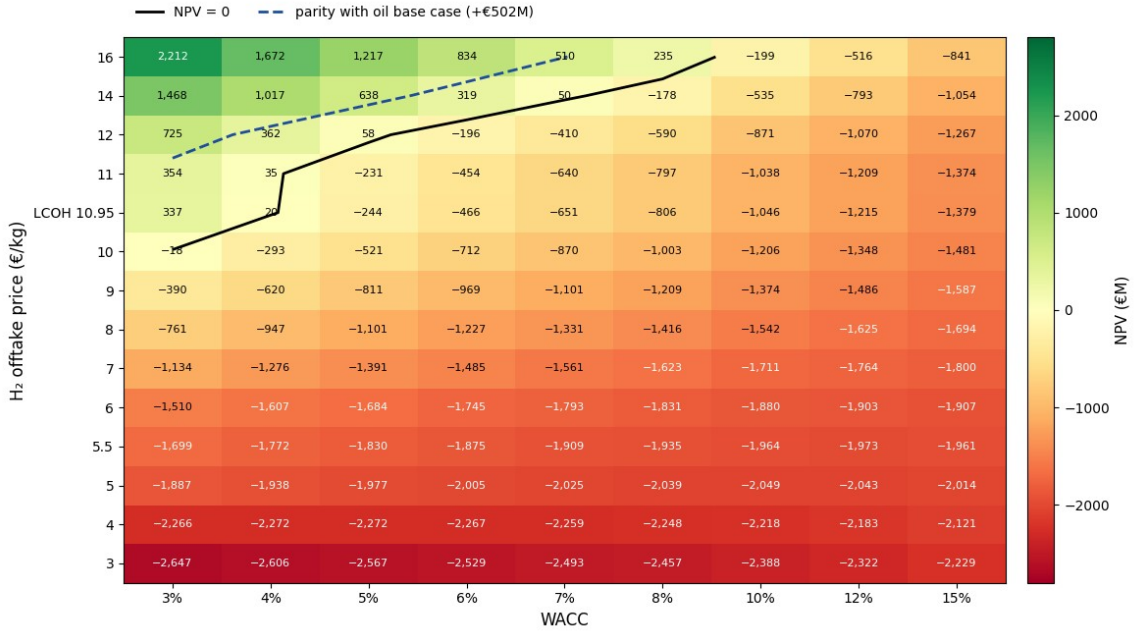
EXPLORATION RISK: A CHEAP, BOUNDED LOTTERY



The gate cuts P(NPV > 0) from 91 % to 27 % and E[NPV] to €71M — still far above hydrogen's near-certain loss (Fig. 5)

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WHEN DOES THE RANKING REVERSE?



Only combined conditions flip the decision: H_2 ≥ €14/kg with WACC < 5 % — or a 65 % petroleum tax on oil (Fig. 6)

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LIMITATIONS — KNOWN DIRECTIONS, ROBUST RANKING

Five qualifications — every one runs against our own conclusion

Oil is an archetype	Bohai costs, mild tax, EU carbon	→ tested: 65 % tax → −€1,444M
Gate ignores abandonment	and compresses timing to year 0	→ understates oil
Scope 3 is accounting	not an equilibrium policy scenario	→ illustrative only
H_2 support excluded	H2Global, CfDs, wind-to-grid sale	→ understates hydrogen
Technical simplifications	parity, uncorrelated MC, PV-factor	→ ranking survived all

None of them overturns the ranking.

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CONCLUSION | A RATIONAL PREFERENCE — BY DESIGN

Rational under current market design — three pillars

1 **96 % of the carbon** is somebody else's
outside the producer's ETS liability

2 **Geological risk is cheap** €80M, bounded, diversifiable
a 30 % shot at a multi-billion upside

3 **Hydrogen has no customer** at a cost-covering price
LCOH €10.96/kg vs. €5.50/kg offtake

Carbon pricing alone will not redirect this capital.

Levers: H_2 offtake contracts · 3–5 % WACC · fossil fiscal design

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THANK YOU

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